

화학수학

여섯 번째 수업

다변수 적분

급수

Vector Derivatives

- Gradient (기울기 연산자) ∇f
scalar $\rightarrow$ vector
- Divergence (발산 연산자) $\nabla \cdot \vec{F}$
vector $\rightarrow$ scalar
- Curl (회전 연산자) $\nabla \times \vec{F}$
vector $\rightarrow$ vector
- Laplacian (라플라스 연산자) $\nabla^2 f$
scalar $\rightarrow$ scalar (2nd order)

Vector Derivatives

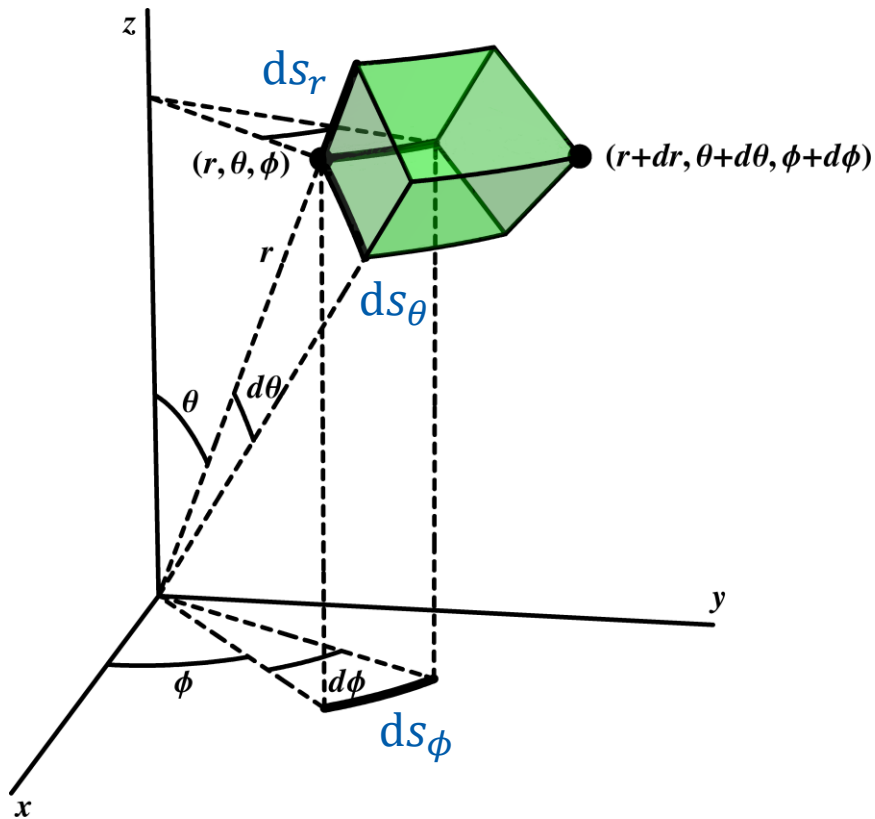
$$\nabla f = \hat{i} \left(\frac{\partial f}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \right)$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right)$$

$$\nabla \times \vec{F} = \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{j} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} \right) + \left(\frac{\partial^2 f}{\partial y^2} \right) + \left(\frac{\partial^2 f}{\partial z^2} \right)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r d\theta$$

$$ds_\phi = r \sin \theta d\phi$$

Infinitesimal Displacements

In Cartesian coordinates,

$$d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

In spherical polar coordinates,

$$d\vec{r} = \hat{e}_r dr + \hat{e}_\theta r d\theta + \hat{e}_\phi r \sin \theta d\phi$$

In general,

$$d\vec{r} = \hat{e}_1 h_1 dq_1 + \hat{e}_2 h_2 dq_2 + \hat{e}_3 h_3 dq_3$$

Vector Derivatives in Other Coordinates

$$\nabla f = \hat{e}_1 \frac{1}{h_1} \frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2} \frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3} \frac{\partial f}{\partial q_3}$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (F_1 h_2 h_3)}{\partial q_1} + \frac{\partial (F_2 h_1 h_3)}{\partial q_2} + \frac{\partial (F_3 h_1 h_2)}{\partial q_3} \right]$$

$$\begin{aligned} \nabla \times \vec{F} = & \hat{e}_1 \frac{1}{h_2 h_3} \left[\frac{\partial (h_3 F_3)}{\partial q_2} - \frac{\partial (h_2 F_2)}{\partial q_3} \right] + \hat{e}_2 \frac{1}{h_1 h_3} \left[\frac{\partial (h_1 F_1)}{\partial q_3} - \frac{\partial (h_3 F_3)}{\partial q_1} \right] \\ & + \hat{e}_3 \frac{1}{h_1 h_2} \left[\frac{\partial (h_2 F_2)}{\partial q_1} - \frac{\partial (h_1 F_1)}{\partial q_2} \right] \end{aligned}$$

Vector Derivatives in Other Coordinates

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Line Integral (선적분)

Consider a differential with two independent variables:

$$dF = M(x, y)dx + N(x, y)dy$$

For the curve C joining (x_0, y_0) and (x_1, y_1) ,
the **line integral** (path integral) is defined as

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

Evaluation of Line Integral

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C dF = \int_C [M\{x, y(x)\}dx + N\{x(y), y\}dy]$$

$$\int_C dF = \int_{x_0}^{x_1} M\{x, y(x)\} dx + \int_{y_0}^{y_1} N\{x(y), y\} dy$$

Line Integrals of Exact Differentials

If dF is an exact differential, the line integral $\int_C dF$

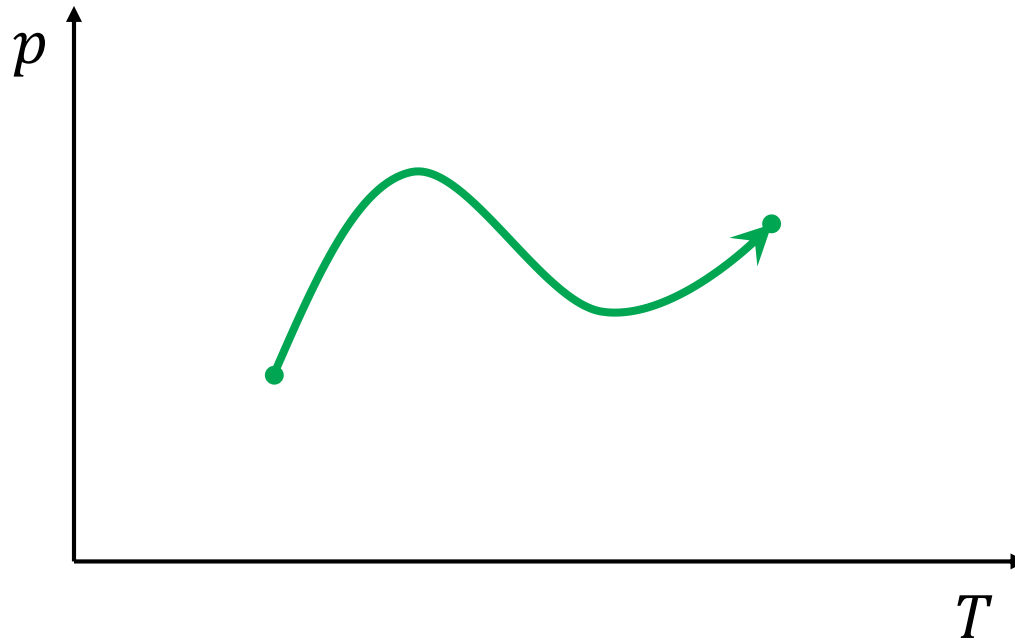
- depends **only on the initial and final points**.
- equals to **$F(\text{final point}) - F(\text{initial point})$** .

For an exact differential dF of the function $F(x, y)$,
with the initial point (x_0, y_0) and the final point (x_1, y_1) ,

$$\int_C dF = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y dx + \left(\frac{\partial F}{\partial y} \right)_x dy \right] = F(x_1, y_1) - F(x_0, y_0)$$

Line Integrals in Thermodynamics

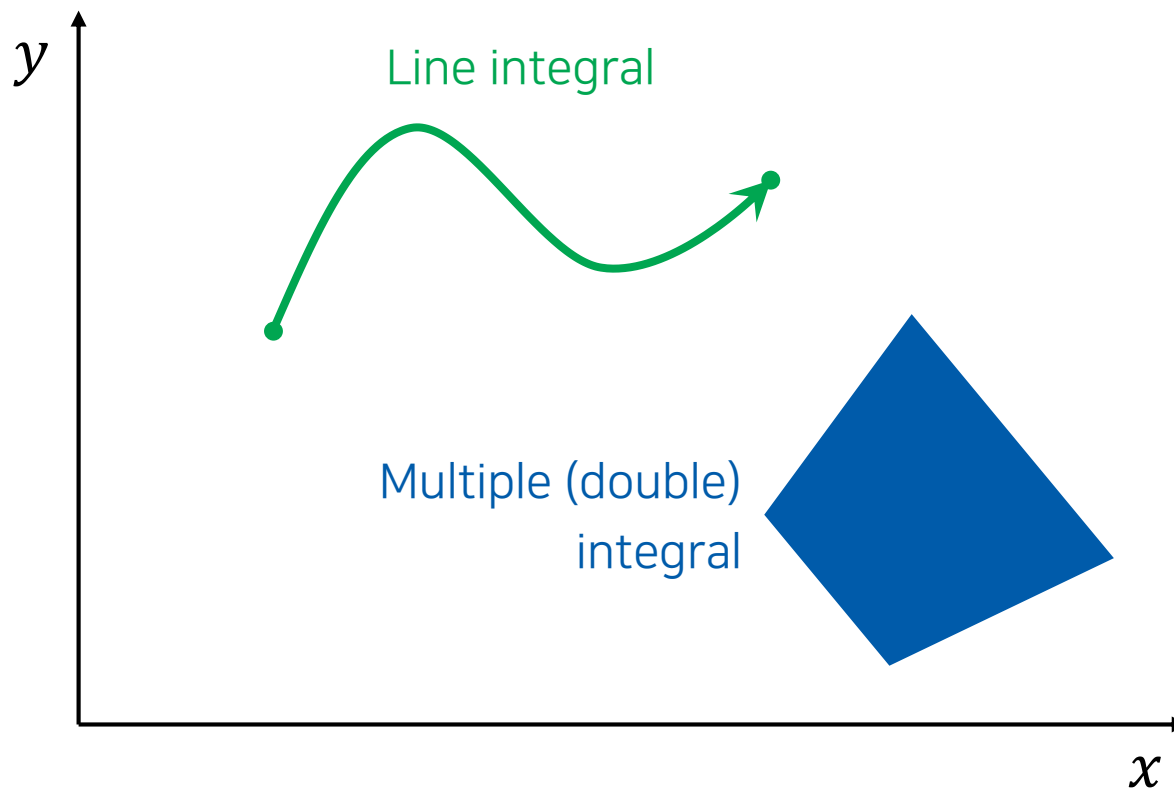
- Equilibrium state: a point in a space
- Reversible process: a line integral



오늘의 수업

- Multiple integrals
 - Double and triple integrals
 - In different coordinate systems
- Series
 - Mathematical series
 - Functional series
 - Power series

Line Integral vs. Multiple Integral



Double Integral (이중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(x, y) \, dy \, dx$$


- Carry out **the inner integration** first
 - Treat the other independent variable (x) as a constant
- Then carry out **the outer integration**

Example 9.6

Evaluate the double integral

$$I = \int_0^a \int_0^b (x^2 + 4xy) \, dy \, dx.$$

Carry out the inner integral first ($x = \text{constant}$):

$$\int_0^b (x^2 + 4xy) \, dy = [x^2y + 2xy^2]_0^b = x^2b + 2xb^2$$

Now carry out the outer integral:

$$I = \int_0^a (x^2b + 2xb^2) \, dx = \left[\frac{b}{3}x^3 + b^2x^2 \right]_0^a = \frac{a^3b}{3} + a^2b^2$$

Example 9.7

Evaluate the double integral

$$\int_0^a \int_0^{3x} (x^2 + 2xy + y^2) \, dy \, dx.$$

Carry out the inner integral first ($x = \text{constant}$):

$$\int_0^{3x} (x^2 + 2xy + y^2) \, dy = \left[x^2 y + xy^2 + \frac{y^3}{3} \right]_0^{3x} = 3x^3 + 9x^3 + 9x^3 = 21x^3$$

Now carry out the outer integral:

$$\int_0^a 21x^3 \, dx = \left[\frac{21}{4} x^4 \right]_0^a = \frac{21}{4} a^4$$

Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

- First, integrate z from c_1 to c_2 ($x, y = \text{constant}$)
- Then integrate y from b_1 to b_2 ($x = \text{constant}$)
- Lastly, integrate x from a_1 to a_2

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz \, dy \, dx.$$

Carry out the innermost integral first ($x, y = \text{constant}$):

$$\int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz = \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \int_0^c \sin^2\left(\frac{k\pi z}{c}\right) dz$$

From the integral table,

$$\int_0^c \sin^2\left(\frac{k\pi z}{c}\right) dz = \frac{c}{2}$$

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

Next, treating x as a constant,

$$\int_0^b \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \frac{c}{2} dy = \sin^2\left(\frac{n\pi x}{a}\right) \frac{c}{2} \int_0^b \sin^2\left(\frac{m\pi y}{b}\right) dy$$

Similarly, from the integral table,

$$\int_0^b \sin^2\left(\frac{m\pi y}{b}\right) dy = \frac{b}{2}$$

Example 9.9

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2\left(\frac{n\pi x}{a}\right) \sin^2\left(\frac{m\pi y}{b}\right) \sin^2\left(\frac{k\pi z}{c}\right) dz dy dx.$$

Lastly,

$$I = A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) \frac{bc}{4} dx = \frac{A^2 bc}{4} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx$$

Similarly, from the integral table,

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2}$$

and we obtain $I = A^2 abc/8$.

Separation of Variables

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

If $f(x, y, z) = f_x(x)f_y(y)f_z(z)$,

$$I = \left[\int_{a_1}^{a_2} f_x(x) \, dx \right] \times \left[\int_{b_1}^{b_2} f_y(y) \, dy \right] \times \left[\int_{c_1}^{c_2} f_z(z) \, dz \right]$$

Example 9.9 Revisited

For integers n , m , and k , find the triple integral

$$I = A^2 \int_0^a \int_0^b \int_0^c \sin^2 \left(\frac{n\pi x}{a} \right) \sin^2 \left(\frac{m\pi y}{b} \right) \sin^2 \left(\frac{k\pi z}{c} \right) dz dy dx.$$

The integral can be decomposed as

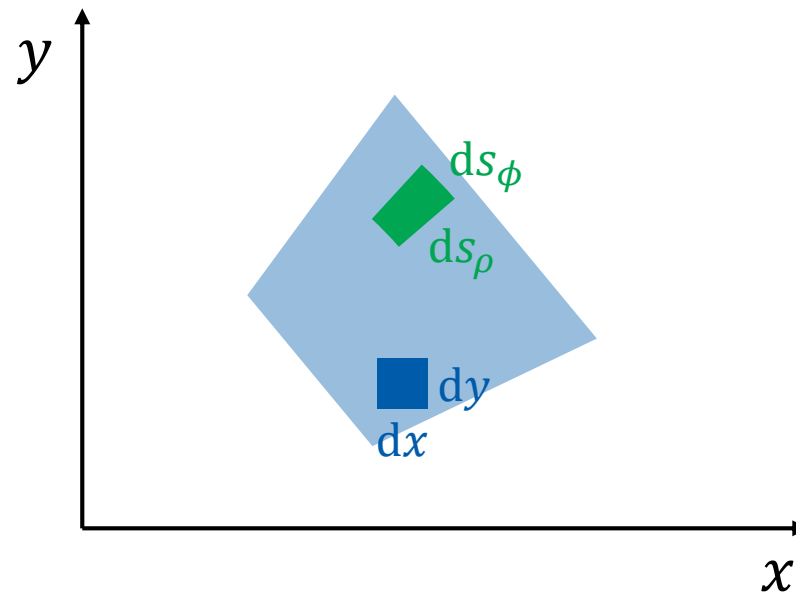
$$I = A^2 \left[\int_0^a \sin^2 \left(\frac{n\pi x}{a} \right) dx \right] \left[\int_0^b \sin^2 \left(\frac{m\pi y}{b} \right) dy \right] \left[\int_0^c \sin^2 \left(\frac{k\pi z}{c} \right) dz \right]$$

Hence, using the integral table,

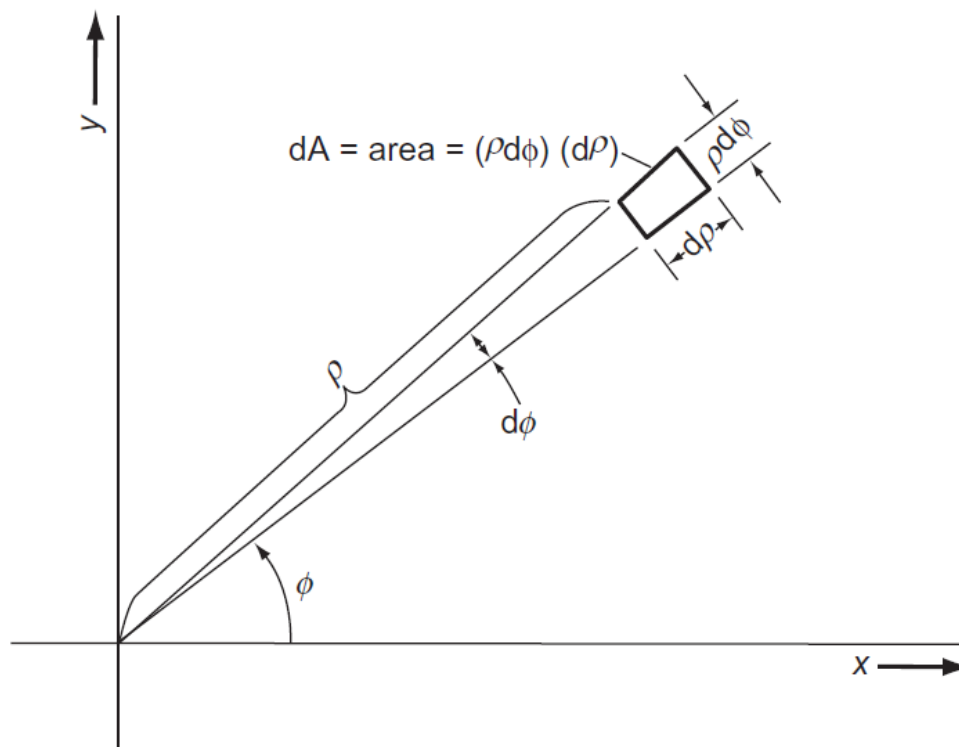
$$I = A^2 \times \frac{a}{2} \times \frac{b}{2} \times \frac{c}{2} = \frac{A^2 abc}{8}$$

Changing Variables in Double Integrals

$$\int \int f(x, y) \, dx \, dy \xrightarrow{?} \int \int f(\rho, \phi) \, d\rho \, d\phi$$



2-Dimensional Polar Coordinates



$$dA = dx dy = \rho d\rho d\phi$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.4)

Conversion Formula

$$\iint f \, dA = \iint f(x, y) \, dx \, dy = \iint f(\rho, \phi) \rho \, d\rho \, d\phi$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

The transformation, $x = \rho \cos \phi$ and $y = \rho \sin \phi$, leads to

$$\begin{aligned}\psi &= B \exp[-a\{(\rho \cos \phi)^2 + (\rho \sin \phi)^2\}] \\ &= B \exp[-a\rho^2(\cos^2 \phi + \sin^2 \phi)] = Be^{-a\rho^2}\end{aligned}$$

Its square is

$$\psi^2 = B^2 e^{-2a\rho^2}$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$\psi^2 = B^2 e^{-2a\rho^2}$$

We require that

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi^2 \, dx \, dy = 1$$

Substitution leads to

$$I = B^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-2a\rho^2} \, dx \, dy$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$\begin{aligned} I &= B^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-2a\rho^2} dx dy \\ &= B^2 \int_0^{2\pi} \int_0^{\infty} e^{-2a\rho^2} \rho d\rho d\phi \\ &= B^2 \int_0^{\infty} e^{-2a\rho^2} \rho d\rho \int_0^{2\pi} d\phi \\ I &= 2\pi B^2 \int_0^{\infty} e^{-2a\rho^2} \rho d\rho \end{aligned}$$

Example 9.10

$$\psi = B \exp[-a(x^2 + y^2)]$$

Transform this function to plane polar coordinates and find the value of B such that the integral of ψ^2 over the entire x - y plane is equal to unity.

$$I = 2\pi B^2 \int_0^\infty e^{-2a\rho^2} \rho \, d\rho$$

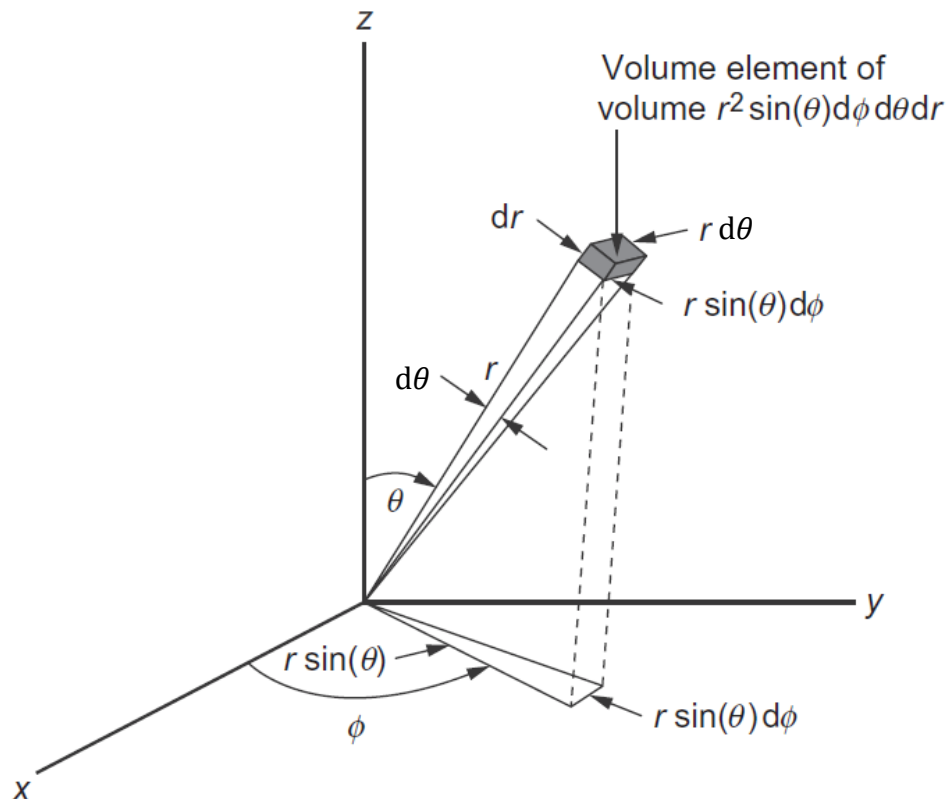
Use the method of substitution, letting $u = 2a\rho^2$ and $du = 4a\rho d\rho$;

$$I = 2\pi B^2 \times \frac{1}{4a} \int_0^\infty e^{-u} \, du = \frac{\pi B^2}{2a}$$

Since $I = 1$, $B^2 = 2a/\pi$ and

$$B = \sqrt{2a/\pi}$$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin \theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Example 9.12

Evaluate the integral

$$I = \int_0^3 \int_0^\pi \int_0^{2\pi} r \cos^2 \phi \, r^2 \sin \theta \, d\phi \, d\theta \, dr.$$

The integral can be decomposed:

$$I = \int_0^3 r^3 \, dr \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} \cos^2 \phi \, d\phi$$

$$\int_0^3 r^3 \, dr = \left[\frac{1}{4} r^4 \right]_0^3 = \frac{81}{4}, \quad \int_0^\pi \sin \theta \, d\theta = [-\cos \theta]_0^\pi = 2$$

$$\int_0^{2\pi} \cos^2 \phi \, d\phi = \int_0^{2\pi} \frac{\cos(2\phi) + 1}{2} \, d\phi = \left[\frac{1}{4} \sin(2\phi) + \frac{1}{2} \phi \right]_0^{2\pi} = \pi$$



Hence, $I = 81\pi/2$.

10. Mathematical Series

(10장 급수)

Sequence and Series

- **Sequence** (수열): $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_n \rightarrow \cdots$
 - Rule for generating one member from previous one
- **Series** (급수): $a_0 + a_1 + a_2 + \cdots$
 - Finite series (유한급수): $s = a_0 + a_1 + \cdots + a_n$
 - Infinite series (무한급수): $s = a_0 + a_1 + \cdots + a_n + \cdots$

More about Infinite Series

$$s = a_0 + a_1 + \cdots + a_n + \cdots = \sum_{k=0}^{\infty} a_k$$

- n th partial sum (부분합): $S_n = a_0 + a_1 + \cdots + a_{n-1}$
 - $s = \lim_{n \rightarrow \infty} S_n$
- If s exists and is finite, the series **converges**; otherwise, the series **diverges**.

Geometric Series (기하급수)

$$s = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = a \sum_{n=0}^{\infty} r^n$$

- Convergence condition: $|r| < 1$
- Value: since $s = a + r(a + ar + ar^2 + \cdots) = a + rs$,

$$s = \frac{a}{1 - r}$$

Example 10.3

The molecular partition function z is defined as

$$z = \sum_{i=0}^{\infty} \exp\left(-\frac{E_i}{k_{\text{B}}T}\right).$$

If we consider only the vibration of a diatomic molecule, to a good approximation,

$$E_i = E_v = h\nu\left(v + \frac{1}{2}\right).$$

Find an expression of the vibrational partition function.

Example 10.3

$$z = \sum_{v=0}^{\infty} \exp\left(-\frac{E_v}{k_B T}\right), \quad E_v = h\nu\left(v + \frac{1}{2}\right)$$

Substitution leads to

$$z = \sum_{v=0}^{\infty} \exp\left[-\frac{h\nu}{k_B T}\left(v + \frac{1}{2}\right)\right] = e^{-h\nu/2k_B T} \sum_{v=0}^{\infty} e^{-h\nu v/k_B T}$$

Introduce $r = e^{-h\nu/k_B T}$ to yield

$$z = r^{1/2} \sum_{v=0}^{\infty} r^v = r^{1/2} \frac{1}{1-r} = \frac{e^{-h\nu/2k_B T}}{1 - e^{-h\nu/k_B T}}$$

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

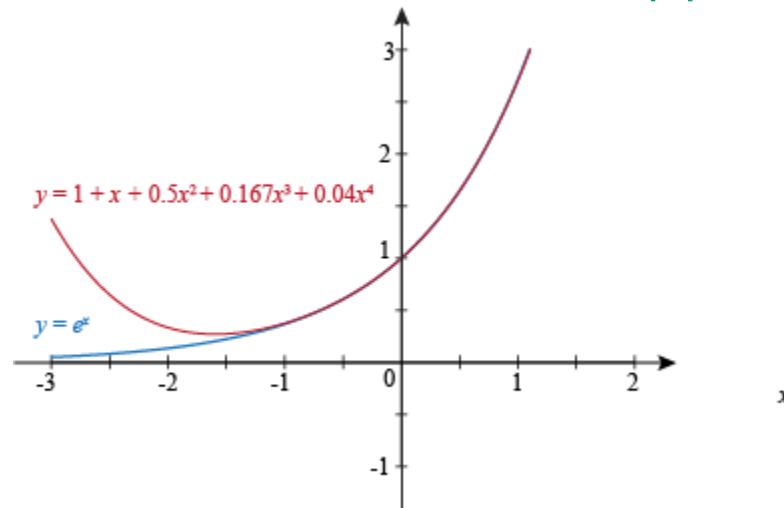
- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Approximation to a Function

- Functional series can be used to represent a function

$$s(x) = a_0g_0(x) + a_1g_1(x) + a_2g_2(x) + \cdots \rightarrow f(x)$$

- Its partial sum can be used as an approximation



(Image: <https://www.shmoop.com/series/taylor-maclaurin-series-examples.html>)

Power Series

- Maclaurin Series

$$s(x) = a_0 + a_1x + a_2x^2 + \dots$$

- Taylor Series

$$s(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

Maclaurin Series to Represent $f(x)$

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

At $x = 0$, $f(0) = a_0$ and

$$\left(\frac{d^n f}{dx^n} \right)_{x=0} = n! a_n$$

Hence,

$$a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

Example 10.7

Find all of the coefficients for the Maclaurin series representing $\sin x$.

Consider $x = 0$. Since $\sin 0 = 0$, $a_0 = 0$.

For $f(x) = \sin x$,

$$\frac{df}{dx} = \cos x, \quad \frac{d^2f}{dx^2} = -\sin x, \quad \frac{d^3f}{dx^3} = -\cos x, \quad \frac{d^4f}{dx^4} = \sin x, \dots$$

Hence, using $a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$,

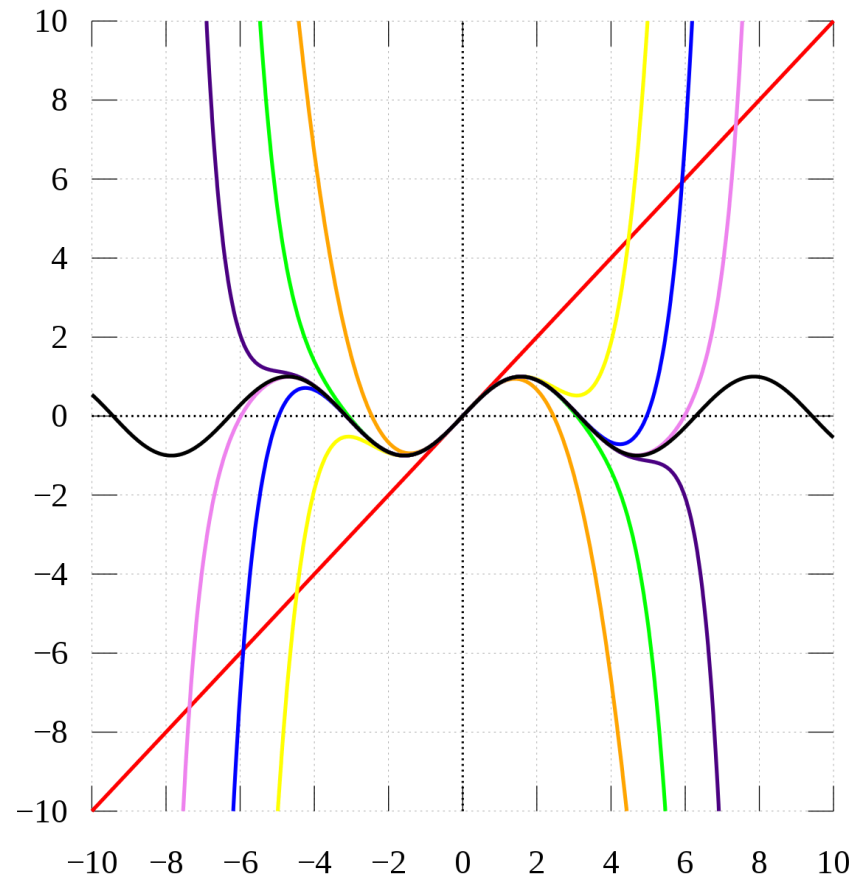
$$a_1 = 1, \quad a_2 = 0, \quad a_3 = -1/3!, \quad a_4 = 0, \dots$$

and

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

Not in Mortimer

Maclaurin Approximation to Sine



(Image: https://commons.wikimedia.org/wiki/File:Sintay_SVG.svg)

Taylor Series to Represent $f(x)$

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

where $h \neq 0$.

$$a_0 = f(h), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=h}$$

- Often called **Taylor expansion about $x = h$**
- A function that is not differentiable at $x = 0$ can be represented by a Taylor series

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

The series is given as

$$\ln x = a_0 + a_1(x - 1) + a_2(x - 1)^2 + a_3(x - 1)^3 + a_4(x - 1)^4 \dots$$

where

$$a_0 = f(1), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=1}$$

For $f(x) = \ln x$, $a_0 = \ln 1 = 0$.

Also,

$$\frac{df}{dx} = \frac{1}{x}, \quad \frac{d^2 f}{dx^2} = -\frac{1}{x^2}, \quad \frac{d^3 f}{dx^3} = \frac{2}{x^3}, \quad \frac{d^4 f}{dx^4} = -\frac{3!}{x^4}, \dots$$

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

$$\frac{df}{dx} = \frac{1}{x}, \quad \frac{d^2f}{dx^2} = -\frac{1}{x^2}, \quad \frac{d^3f}{dx^3} = \frac{2}{x^3}, \quad \frac{d^4f}{dx^4} = -\frac{3!}{x^4}, \dots$$

Hence,

$$\begin{aligned} a_1 &= \frac{1}{1!} \left(\frac{df}{dx} \right)_{x=1} = 1, & a_2 &= \frac{1}{2!} \left(\frac{d^2f}{dx^2} \right)_{x=1} = -\frac{1}{2}, \\ a_3 &= \frac{1}{3!} \left(\frac{d^3f}{dx^3} \right)_{x=1} = \frac{1}{3}, & a_4 &= \frac{1}{4!} \left(\frac{d^4f}{dx^4} \right)_{x=1} = -\frac{1}{4}, \dots \end{aligned}$$

Therefore,

$$\ln x = (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3 - \frac{1}{4}(x - 1)^4 + \dots$$

Change of Variable

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

can be written as

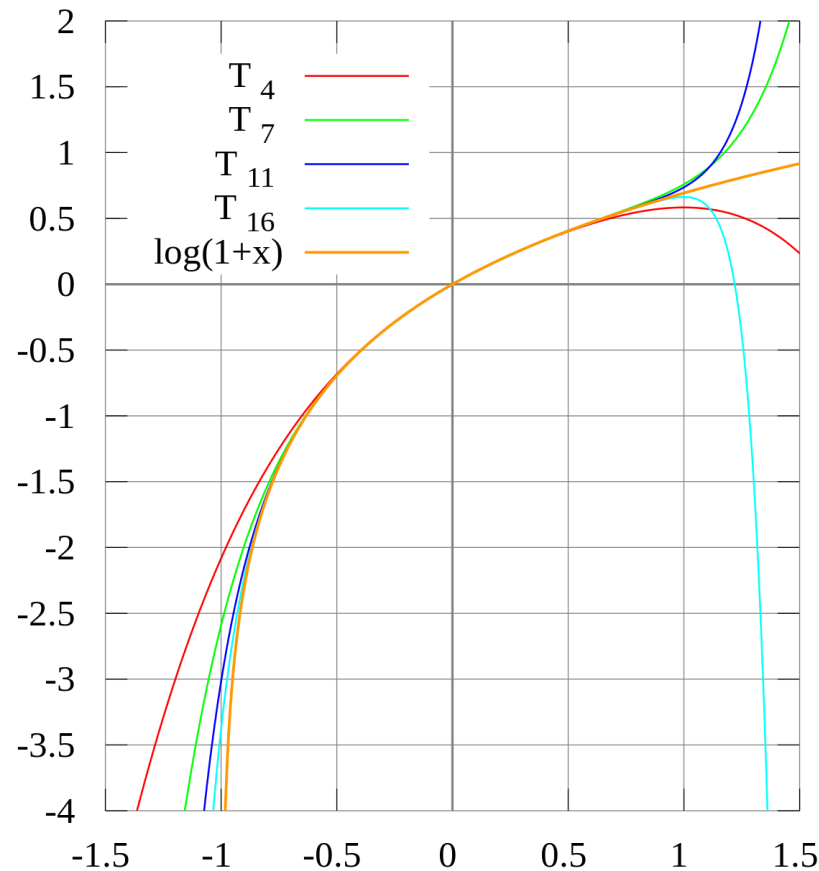
$$f(x + h) = a_0 + a_1x + a_2x^2 + \dots$$

$$a_0 = f(x = 0), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

e.g. Taylor expansion of $\ln(x + 1)$:

$$\ln(x + 1) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

Taylor Approximation to $\ln(x + 1)$



(Image: <https://commons.wikimedia.org/wiki/File:LogTay.svg>)

Mathematical Operations on Series

Differentiation/integration of a series?

For a (uniformly convergent) functional series

$$f(x) = \sum_{n=0}^{\infty} a_n g_n(x),$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] = \sum_{n=0}^{\infty} a_n \frac{dg_n(x)}{dx}$$

$$\int_b^c f(x) dx = \int_b^c \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] dx = \sum_{n=0}^{\infty} a_n \int_b^c g_n(x) dx$$

Example 10.12

Find the Maclaurin series for $\cos x$ from the Maclaurin series for $\sin x$.

The Maclaurin series for $\sin x$ is

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

Since $d(\sin x)/dx = \cos x$, differentiating both sides,

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ + a_{21}x^2y + a_{12}xy^2 + \dots$$

$$x = 0, y = 0 \text{ gives } f(0, 0) = a_{00}$$

What about other coefficients?

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ + a_{21}x^2y + a_{12}xy^2 + \dots$$

$$(\partial f / \partial x)_y = a_{10} + a_{11}y + 2a_{20}x + 2a_{21}xy + a_{12}y^2 + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial f / \partial x)_y|_{0,0} = a_{10}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$\begin{aligned} f(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + \dots \end{aligned}$$

$$(\partial f / \partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial f / \partial y)_x|_{0,0} = a_{01}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$\begin{aligned} f(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + \dots \end{aligned}$$

$$(\partial f / \partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \dots$$

$$(\partial^2 f / \partial x \partial y) = a_{11} + 2a_{21}x + 2a_{12}y + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial^2 f / \partial x \partial y)|_{0,0} = a_{11}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ + a_{21}x^2y + a_{12}xy^2 + \dots$$

In general,

$$a_{nm} = \frac{1}{n! m!} \left(\frac{\partial^{n+m} f}{\partial^m y \partial^n x} \right) \Big|_{0,0}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x, y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y .

Consider $f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + \dots$

$$f(0,0) = e^0 = 1 = a_{00}$$

$$(\partial f / \partial x)_y \Big|_{0,0} = e^{x+y} \Big|_{0,0} = e^0 = 1 = a_{10}$$

$$(\partial f / \partial y)_x \Big|_{0,0} = e^{x+y} \Big|_{0,0} = e^0 = 1 = a_{01}$$

All the derivatives are equal to e^{x+y} and

$$a_{nm} = \frac{1}{n! m!} \left(\frac{\partial^{n+m} f}{\partial^m y \partial^n x} \right) \Big|_{0,0} = \frac{1}{n! m!}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x, y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y .

$$e^{x+y} = 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \dots$$

The Maclaurin series representing e^x is

$$e^x = 1 + x + (1/2!)x^2 + (1/3!)x^3 + \dots$$

Similarly,

$$e^y = 1 + y + (1/2!)y^2 + (1/3!)y^3 + \dots$$

Hence, the product of e^x and e^y is

$$\begin{aligned} e^{x+y} &= (1 + x + (1/2!)x^2 + \dots)(1 + y + (1/2!)y^2 + \dots) \\ &= 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \dots \end{aligned}$$